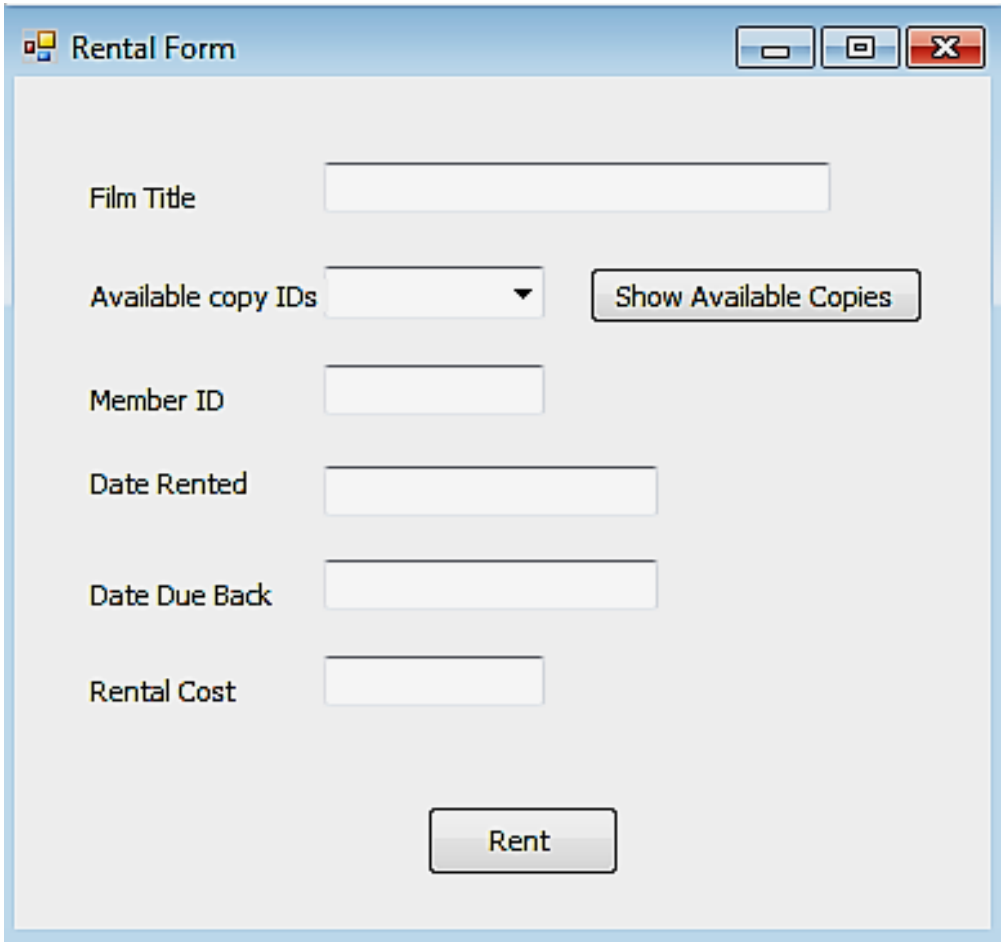
Software Engineering Lab

Building .NET Applications on the Oracle Database with Microsoft Visual Studio

Lab 3 – ODP.Net
Oracle Stored Procedures connected layer

Inserting New Rental Information using Stored Procedure



What Is a Procedure?

- Is a named PL/SQL block that can accept parameters and be invoked from a calling environment to perform an action.
- A procedure can be stored in the database, as a schema object, for repeated execution.

Syntax for Creating Procedures

```
CREATE [OR REPLACE] PROCEDURE procedure_name
 [(parameter1 [mode] datatype1,
   parameter2 [mode] datatype2, ...)]
IS|AS
  [local_variable_declarations; …]
BEGIN
  -- actions;
END [procedure_name];
```

PL/SQL Block

Procedural Parameter Modes

- Select one of the three modes for each parameter:
- *IN (default):* Passes a constant value from the calling environment into the procedure
- *OUT:* Passes a value from the procedure to the calling environment
- *IN OUT:* parameter supplies an input value, which may be returned (output) as a modified value

Calling Procedures From .Net Application

A- using SQL Developer

- create the stored Procedure

B- using .Net Application

- OracleConnection
- OracleCommand
  1. Connection
  2. CommandText → procedure name
  3. CommandType → storedprocedure
  4. Parameters → In/Out
  5. Execute Command

Handling SQL statements inside Stored Procedure

SQL statement	In Parameter	Out Parameter	Calling Procedure from .Net
Select Single row	Depends on the question (i.e. when input info from .Net should be passed to procedure)	Define output parameter for each column in the select statement	ExecuteNonQuery
Select Multiple rows		Define one output parameter with data type: Sys_refCursor	ExecuteReader
Insert/update/ delete		Depends on the question	ExecuteNonQuery

Handling Select Statement inside stored procedure

- If the procedure returns result set with :

  - Single Row: Create the PL/SQL block in procedure with *out* parameters

    SELECT *columns* INTO *output parameters*

  - Multiple Rows: Create the PL/SQL block the procedure with *sys_refcursor* parameter

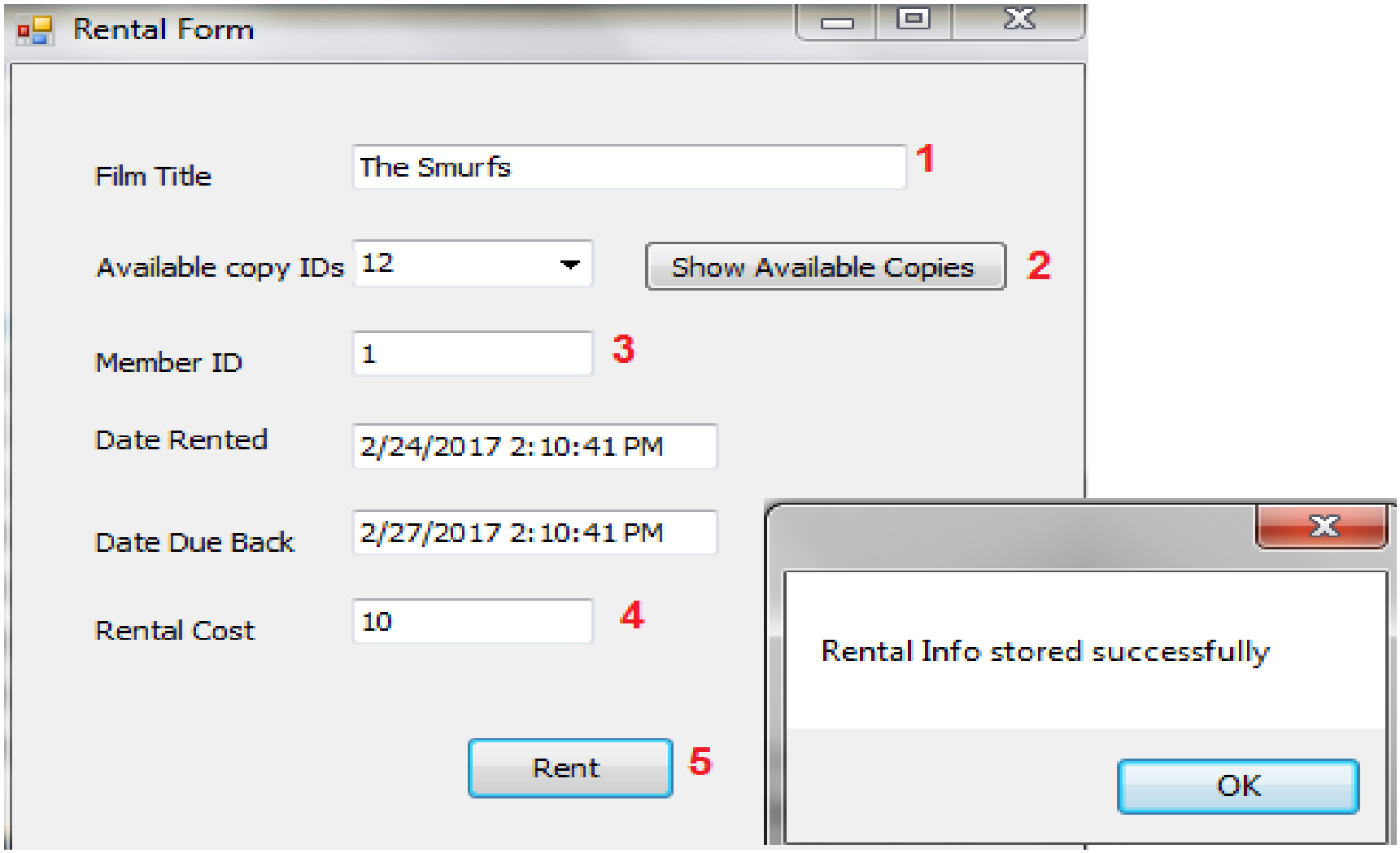
    OPEN *refcursor_name* for

    select statement

Sys_refcursor Parameter

- A cursor variable that is passed as a parameter to stored procedure with a multi-row query.

Rental form


Rental Form
Film Title
The Smurfs
1
Available copy IDs
12
Show Available Copies
2
Member ID
1
3
Date Rented
2/24/2017 2:10:41 PM
Date Due Back
2/27/2017 2:10:41 PM
Rental Cost
10
4
Rent
5
Rental Info stored successfully
OK

Implementation Steps

1- Loading FilmCopies ComboBox with IDs of Available copies in showCopies_click event

2- Set DateRented with today's date

3- Set Due Date with three days from the date the DVD is rented

4- To Insert a new rental, we must pass:

- RentID, MemberID, FilmCopyID, DateRented, Due Date, Rental Cost to Rental Table

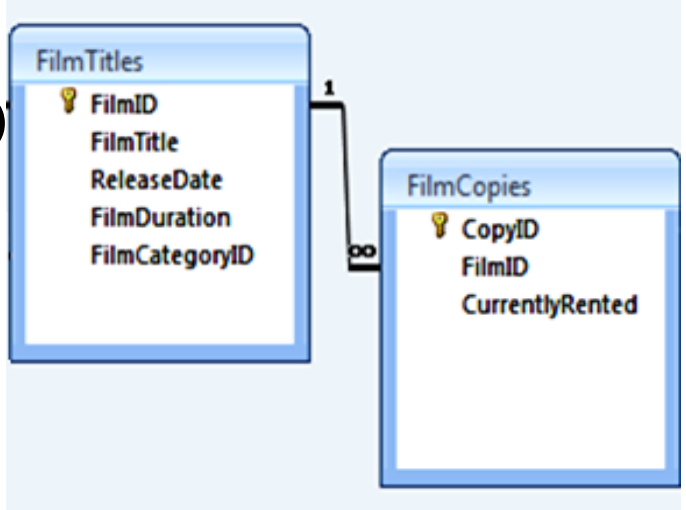
- So we must store New value for RentID

5- Set the film status to RENTED

1- Loading FilmCopies ComboBox with Copy IDs

Call GetCopyID Procedure To Get Film Copy ID

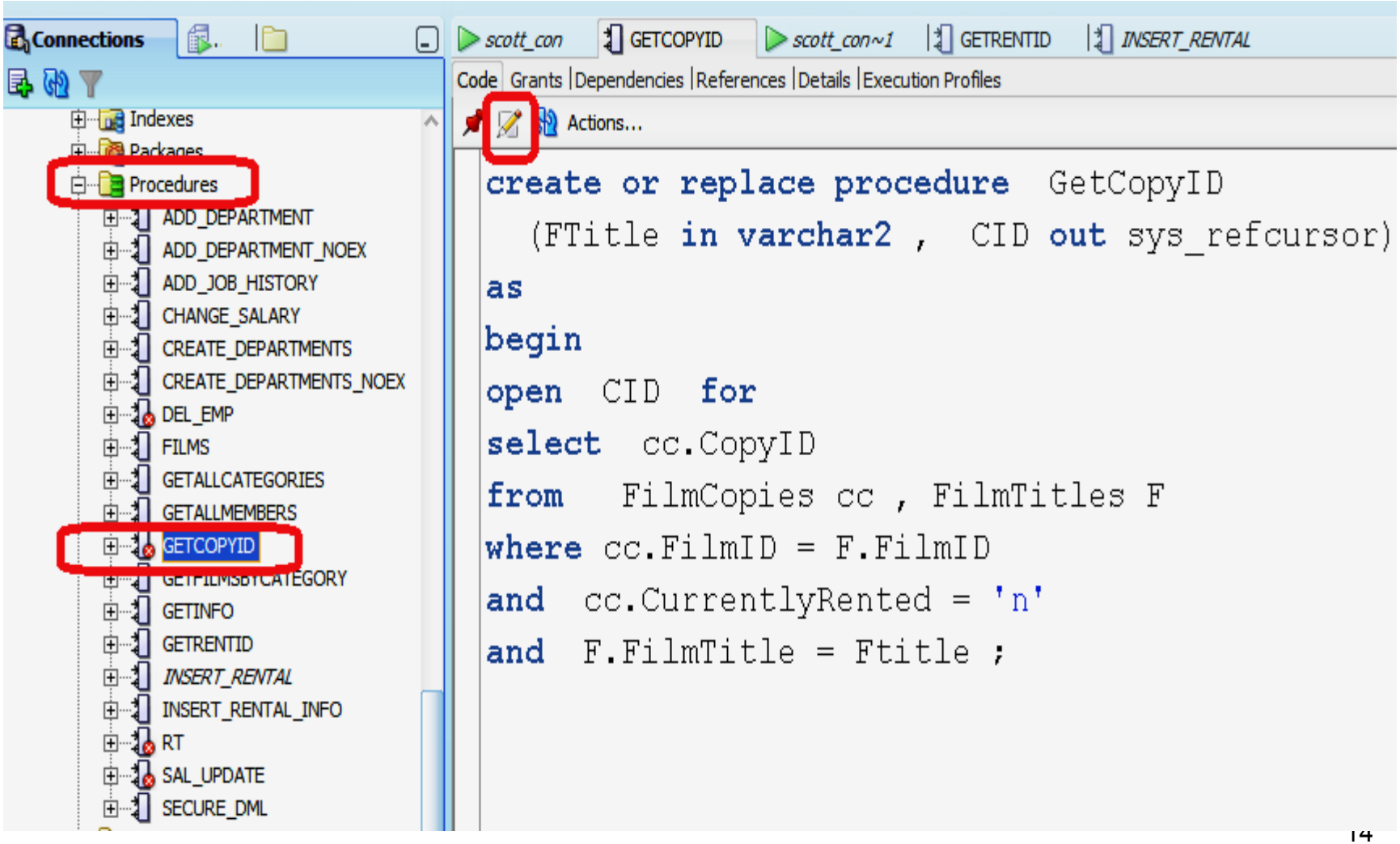
```
create or replace procedure  GetCopyID
(FTitle in varchar2 ,  CID out sys_refcursor)
as
begin
open  CID  for
select  fc.CopyID
from   FilmCopies fc , FilmTitles F
where fc.FilmID = F.FilmID  and  fc.CurrentlyRented = 'n'  and
          F.FilmTitle = Ftitle ;
end;
```



Procedure Compilation

- Successful:
  procedure *procedureName* Compiled.
- Failed:
  Warning: execution completed with warning
  procedure *procedureName* Compiled.

Edit Procedure Definition



Edit Procedure and Correct Errors

Connections

- Indexes
- Packages
- Procedures
  - ADD_DEPARTMENT
  - ADD_DEPARTMENT_NOEX
  - ADD_JOB_HISTORY
  - CHANGE_SALARY
  - CREATE_DEPARTMENTS
  - CREATE_DEPARTMENTS_NOEX
  - DEL_EMP
  - FILMS
  - GETALLCATEGORIES
  - GETALLMEMBERS
  - GETCOPYID
  - GETFILMSBYCATEGORY
  - GETINFO
  - GETRENTID
  - *INSERT_RENTAL*
  - INSERT_RENTAL_INFO
  - RT
  - SAL_UPDATE
  - SECURE_DML
- Functions
- Queues
- Queues Tables
- Triggers
- Types

scott_con | GETCOPYID | GETCOPYID | scott_con~1 | GETRENTID | *INSERT_RENTAL*

```
create or replace
procedure  GetCopyID
   (FTitle in varchar2 ,  CID out sys_refcursor)
as
begin
open  CID  for
select  cc.CopyID
from   FilmCopies cc , FilmTitles F
where cc.FilmID = F.FilmID
and  cc.CurrentlyRented = 'n'
and  F.FilmTitle = Ftitle ;
```

End; is missing

Project: C:\Users\Naglaa\AppData\Roaming\SQL Developer\system1.5.5.59.69\o.sqldeveloper.11.1.1.59.69\DefaultWorkspace

PROCEDURE SCOTT.GETCOPYID@scott_con

Error(12): PLS-00103: Encountered the symbol "end-of-file" when expecting one of the following: (begin case declar

Creating Global Oracle Connection Object in .Net App

```
public partial class Rental_Form : Form
{
    OracleConnection con;
    string ordb = "data source = orcl; user id =scott; password=tiger;";

    public Rental_Form()
    {
        InitializeComponent();
    }

    private void Rental_Form_Load(object sender, EventArgs e)
    {
        con = new OracleConnection(ordb);
        con.Open();

    }
```

1- Loading FilmCopies ComboBox with Copy IDs in button_click event

- Calling GetCopyID From .Net App:

- Create OracleCommand Object:

```
OracleCommand cmd = new OracleCommand();
cmd.Connection = con;
```

- Set CommandText property to the name of the stored procedure:

```
cmd.CommandText = "GetCopyID";
```

1- Loading FilmCopies ComboBox with Copy IDs in button_click event

- Set the CommandType property to indicate that it is a stored procedure :

```
cmd.CommandType = CommandType.StoredProcedure;
```

- Add Command Parameters that represent:
  - Input Parameter

```
cmd.Parameters.Add("title", txt_title.Text);
```

  - the ref cursor output parameter

```
cmd.Parameters.Add("id", OracleDbType.RefCursor,
ParameterDirection.Output);
```

1- Loading FilmCopies ComboBox with Copy IDs in button_click event

- Execute the command

```
OracleDataReader dr = cmd.ExecuteReader();
```

- The result set (rows) will be returned in the RefCursor output parameter.
- We can not directly fetch rows from cursor parameter.
- Instead, we declare OracleDataReader object that will be implicitly linked to the result set (rows) returned into the RefCursor output parameter.

1- Loading FilmCopies ComboBox with Copy IDs in button_click event

- Calling GetCopyID From .Net App:

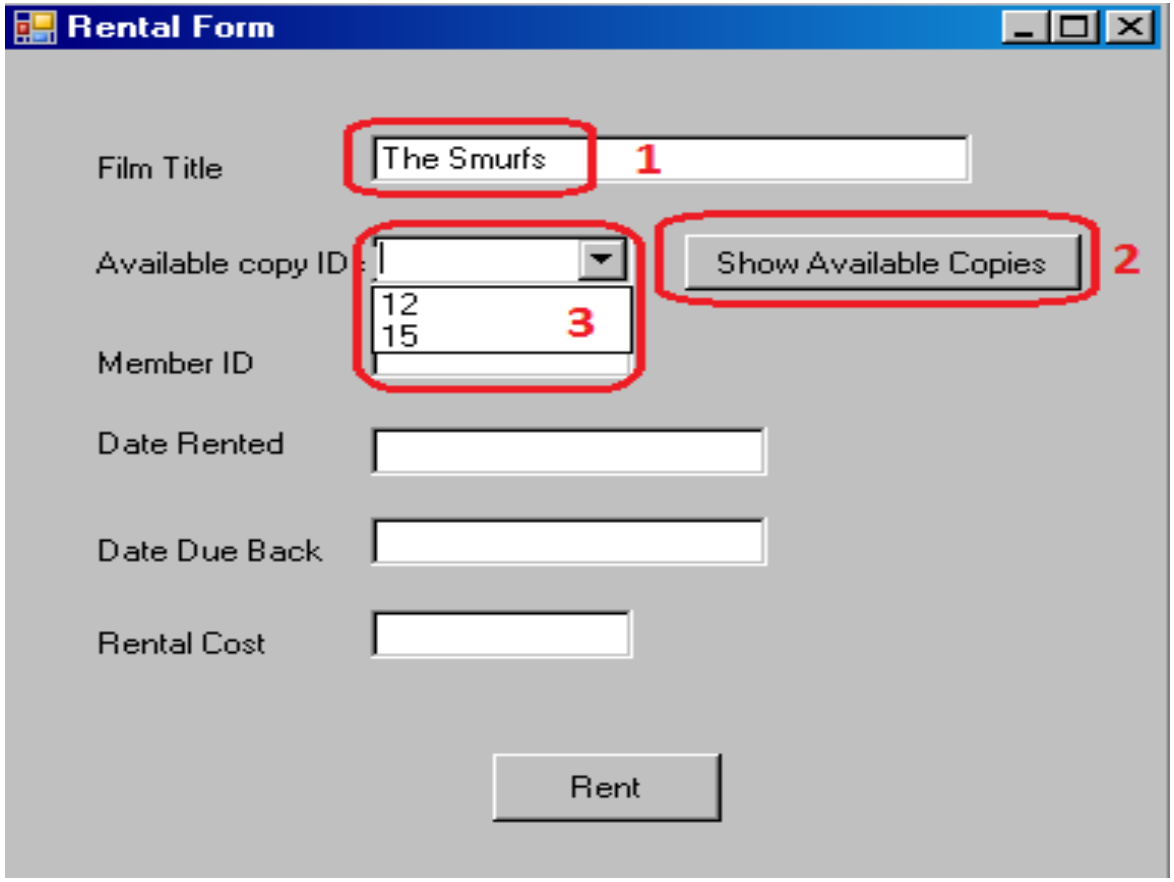
```
private void btn_showCopies_Click(object sender, EventArgs e)
{
    OracleCommand cmd = new OracleCommand();
    cmd.Connection = con;
    cmd.CommandText = "GetCopyID";
    cmd.CommandType = CommandType.StoredProcedure;
    cmd.Parameters.Add("title", txt_title.Text);
    cmd.Parameters.Add("id", OracleDbType.RefCursor, ParameterDirection.Output);

    OracleDataReader dr = cmd.ExecuteReader();
    while (dr.Read())
    { cmb_copies.Items.Add(dr[0]); }

   dr.Close();
}
```

Execution & Retrieving the results

1- Loading FilmCopies ComboBox with Copy IDs in button_click event

Film title Example:

The Smurfs



2- Set DateRented with today's date

- Using DateTime.Now Property:

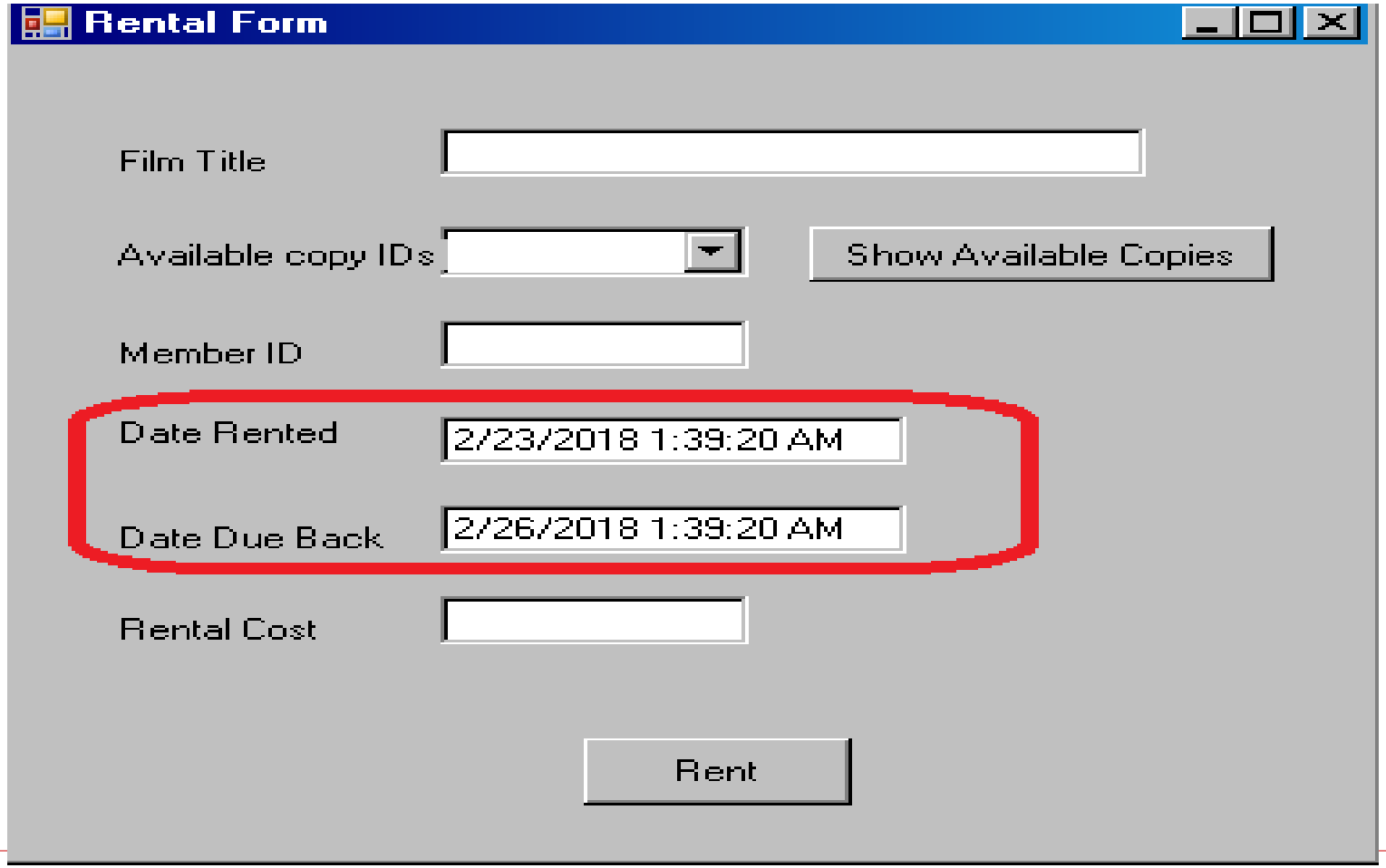
```
private void Rental_Form_Load(object sender, EventArgs e)
{
    con = new OracleConnection(ordb);
    con.Open();

    txt_DateRented.Text = DateTime.Now.ToString();

    txt_DateBack.Text = DateTime.Now.AddDays(3).ToString();

}
```

3- Set Due Date with three days from the date the DVD is rented

- Using AddDays() Function:

```
private void Rental_Form_Load(object sender, EventArgs e)
{
    con = new OracleConnection(ordb);
    con.Open();

    txt_DateRented.Text = DateTime.Now.ToString();

    txt_DateBack.Text = DateTime.Now.AddDays(3).ToString();

}
```

Set values for Date rented & Date_Due_Back



Thank You